Experiment 8

Aim: Perform word sense disambiguation using WordNet

Theory:

Text summarization is the process of condensing a large amount of text into a shorter version while preserving the key information and overall meaning. It helps in quickly understanding the important points of a document without reading the entire content.

WordNet is a lexical database that provides semantic relationships between words such as synonyms, hypernyms, and related meanings. In text summarization, WordNet is used to understand the importance of words based on their meanings and relationships. Words that are semantically related and frequently occurring in the document are considered more significant.

The process begins with preprocessing steps such as tokenization, removal of stop words, and normalization. After preprocessing, WordNet is used to analyze semantic similarity between words and sentences. Based on this analysis, scores are assigned to sentences depending on the importance of the words they contain. Sentences with higher scores are selected and arranged to form a meaningful summary.

This approach helps generate summaries that retain the main ideas of the original text and is useful in applications like document summarization, news summarization, and information retrieval.

Source Code and Outputs:

```
!pip install nltk

import nltk
import numpy as np
from nltk.corpus import stopwords, wordnet as wn
from nltk.tokenize import word_tokenize, sent_tokenize
from collections import defaultdict
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
True
```

```python
text = """
Artificial intelligence is transforming industries across the world.
It enables machines to learn from data and make intelligent decisions.
AI is widely used in healthcare, finance, and autonomous vehicles.
However, it also raises ethical concerns such as job loss and bias.
Researchers are working to make AI more transparent and fair.
"""

print(text)
```

```
Artificial intelligence is transforming industries across the world.
It enables machines to learn from data and make intelligent decisions.
AI is widely used in healthcare, finance, and autonomous vehicles.
However, it also raises ethical concerns such as job loss and bias.
Researchers are working to make AI more transparent and fair.
```

```python
sentences = sent_tokenize(text)

stop_words = set(stopwords.words('english'))

word_frequencies = defaultdict(int)

for sentence in sentences:
    words = word_tokenize(sentence.lower())
    for word in words:
        if word.isalnum() and word not in stop_words:
            word_frequencies[word] += 1

# Normalize frequencies
max_freq = max(word_frequencies.values())

for word in word_frequencies:
    word_frequencies[word] /= max_freq

print(word_frequencies)
```

```
defaultdict(<class 'int'>, {'artificial': 0.5, 'intelligence': 0.5, 'transforming': 0.5, 'industries': 0.5, 'across': 0.5, 'world': 0.5, 'enables': 0.5
```

```python
def get_synonyms(word):
    synonyms = set()
    for syn in wn.synsets(word):
        for lemma in syn.lemmas():
            synonyms.add(lemma.name())
    return synonyms
```

```python
sentence_scores = defaultdict(int)

for sentence in sentences:
    words = word_tokenize(sentence.lower())

    for word in words:
        if word in word_frequencies:
            sentence_scores[sentence] += word_frequencies[word]

            # Add score for synonyms (WordNet)
            synonyms = get_synonyms(word)
            for syn in synonyms:
                if syn in word_frequencies:
                    sentence_scores[sentence] += word_frequencies[syn]
```

```python
# Select top 2 sentences
import heapq

summary_sentences = heapq.nlargest(2, sentence_scores, key=sentence_scores.get)

summary = ' '.join(summary_sentences)

print("Summary:")
print(summary)
```

```
Summary:
Researchers are working to make AI more transparent and fair. AI is widely used in healthcare, finance, and autonomous vehicles.
```

Learning Outcomes:

I am able to understand and implement text summarization techniques using WordNet by identifying key concepts, removing redundant information, and generating concise summaries that retain the essential meaning of the original text, making large documents easier to process and analyze.